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Radiative Transfer Modeling of Circumstellar Disks via RADMC-3D Simulations

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1 Introduction

When a young star forms from the collapse of molecular clouds, they have a dense, rotating Protoplanetary Disk attached to it. This disk is formed out of the conservation of angular momentum in star formation. They represent the environment in which planets form, grow, and migrate and any understanding of planet formation is intimately coupled to understanding the protoplanetary disk.

A colour-colour diagram of a range of young stellar objects shows two major classifications: young stars with an optically thick protoplanetary disk (marked by excess NIR) and YSOs with a very consistent stellar photosphere indicating that they are older and have lost their disk. However, a subset of objects does not fall clearly into either the primordial or disc-less categories. These sources exhibit little to no near-infrared (NIR) excess, yet show mid-infrared (MIR) excesses consistent with those of primordial disks. This suggests a lack of disk emission from the inner regions, while retaining optically thick emission at larger radii. Because these objects occupy an intermediate position in colour-colour space between primordial disks and disk-less stars, they are thought to represent systems undergoing disk dispersal. Consequently, such disks are interpreted as being in a transitional evolutionary phase and are referred to as **transition disks** [1].

Several theories suggest the formation of Transition disks from a Protoplanetary disk. The current and most widely accepted theory is formation through pebble accretion. A system of multiple low-mass planets can act as an efficient dust filtration mechanism, wherein the planets accrete inward-drifting pebbles and inhibit their transport to the inner disc regions. This process leads to a depletion of small solids interior to the planetary orbits, thereby producing an inner cavity [2].

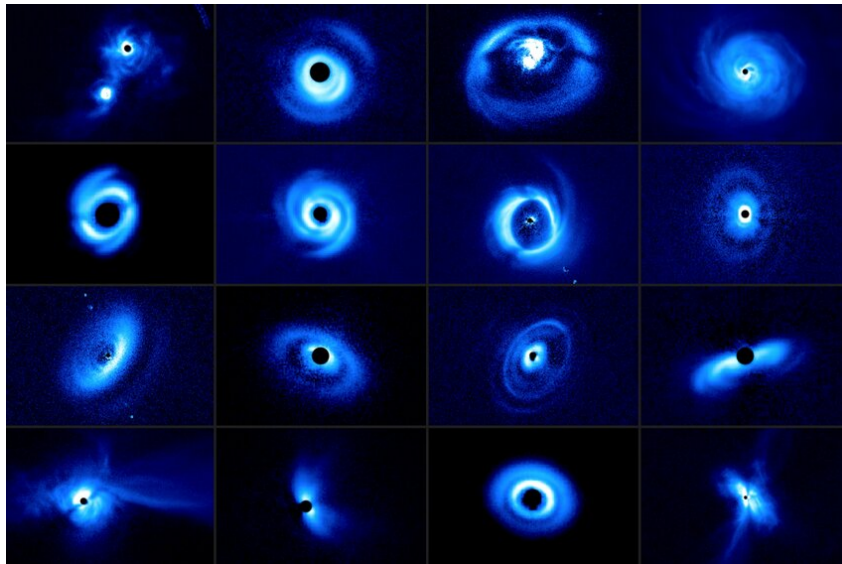


Figure 1: Observations from VLT/SPHERE showing diversity of disk structures, including spiral arms, rings, and inner cavities characteristic of the transitional phase

The structure of a transition disk is defined by a distinct spatial decoupling between its gas and dust components. While the system appears "hollowed out" in the infrared due to a massive central cavity depleted of millimeter-sized dust, high-resolution ALMA

observations reveal that this gap is often a selective filter. Gas and very fine grains frequently persist within the cavity, forming a small, residual inner disk that continues to facilitate stellar accretion. This architecture—consisting of a compact inner disk, a wide depleted gap, and a massive outer dust ring—is primarily shaped by the pressure gradients of hidden protoplanets, which trap larger solids in the outer regions while allowing gas to stream inward toward the host star [3].

2 Computational Methodology - RADMC-3D

In our project, we used the software, **RACMC-3D**[\[4\]](#), developed by C.P. Dullemond, to post-process models for Protoplanetary disks and Transition Disks.

RADMC-3D is a versatile radiative transfer code used to generate synthetic observables—such as spectra, spectral energy distributions (SEDs), and images—from astrophysical models. Rather than being a physical model itself, it serves as a post-processing tool in which users define the underlying structure, including the spatial grid, density distribution of dust and gas, and radiation sources.

The code computes dust continuum radiative transfer using a Monte Carlo approach, enabling self-consistent determination of dust temperatures, while line transfer can be treated under LTE or approximate non-LTE conditions (e.g., LVG). Its highly flexible framework allows users to input custom microphysics, such as dust opacities and molecular data, through external files. RADMC-3D is particularly well suited for post-processing outputs from (magneto-)hydrodynamic simulations. Although its computational core is written in Fortran 90, it is typically interfaced through file-based workflows and supported by the Python package `radmc3dPy`, which facilitates data handling and analysis.

3 Simulated Results and Parametric Analysis of Disk Properties

We start by studying and understanding the structure of a protoplanetary disk, by running an example `run_ppdisk_analytic_1`.

3.1 Baseline Model Characterization- modeling a simple disk

Having RADMC-3D installed, in its terminal, open the directory with our disk model (our examples in this case) run the following step by step:

1. `problem_setup.py`
2. `radmc3d mctherm`
3. `radmc3d image`
4. `radmc3d sed`

3.1.1 `problem_setup.py`

It creates all the necessary `.inp` (input) files. In our problem, we required the following input files:

1. `radmc3d.inp` : Contains all the main settings of RADMC-3D. An empty file means all settings are default. We only needed to alter these values:
 - `nphot = 1000000`, i.e. number of photons
 - `scattering_mode_max = 1`, i.e. isotropic scattering
 - `iranfreqmode = 1`
2. `stars.inp` : This specifies the number of stars, their position, radii, mass and spectra. The file is written in a specific ASCII format.
3. `amr_grid.inp` : This specifies the spectral grid of the model. It can be of three types: (i) Regular grid, (ii) Oct tree style AMR, (iii) Layer style AMR. We have a regular grid with size of grid and all three cell walls specified.
4. `dust_density.inp` : This creates a grid of dust density values using the `amr_grid` specifying the number of cells and dust species.
5. `dustopac.inp` : This specifies the dust opacities and tells us which dust species to read.
6. `wavelength_micron.inp` : It sets the discrete wavelength points for continuum radiative transfer calculation. Several wavelength values have been concatenated to `lam` and its size `nlam` stands for number of frequency points for stellar spectra.

3.1.2 `radmc3d mctherm`

Once the input files are created, this step tells RADMC-3D to run the thermal Monte Carlo simulation. The `dust_temperature.out` is created. It creates a grid of temperature of the dust in the entire star and disk system. Visualising it in several layers, we see

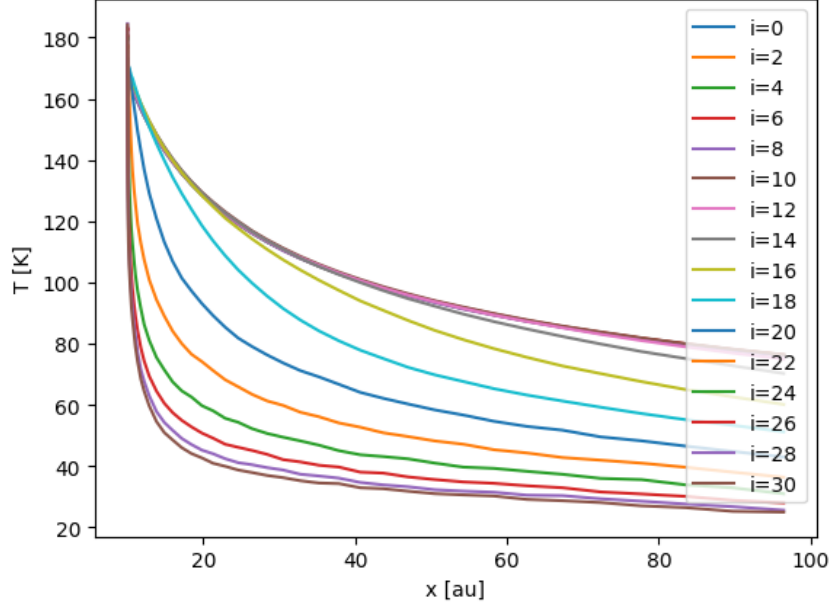


Figure 2: Dust temperature distribution in the protoplanetary disk computed via Monte Carlo radiative transfer with RADMC-3D.

3.1.3 radmc3d image lambda 10 incl 45 phi 30 secondorder

We use `image` to generate a synthetic image of the star-disk system, with the output stored in `image.out`. The image is computed at a wavelength of $10\,\mu\text{m}$, viewed at an inclination of 45° from the z-axis and an azimuthal rotation of 30° about the z-axis, with second-order integration applied for smoothing.

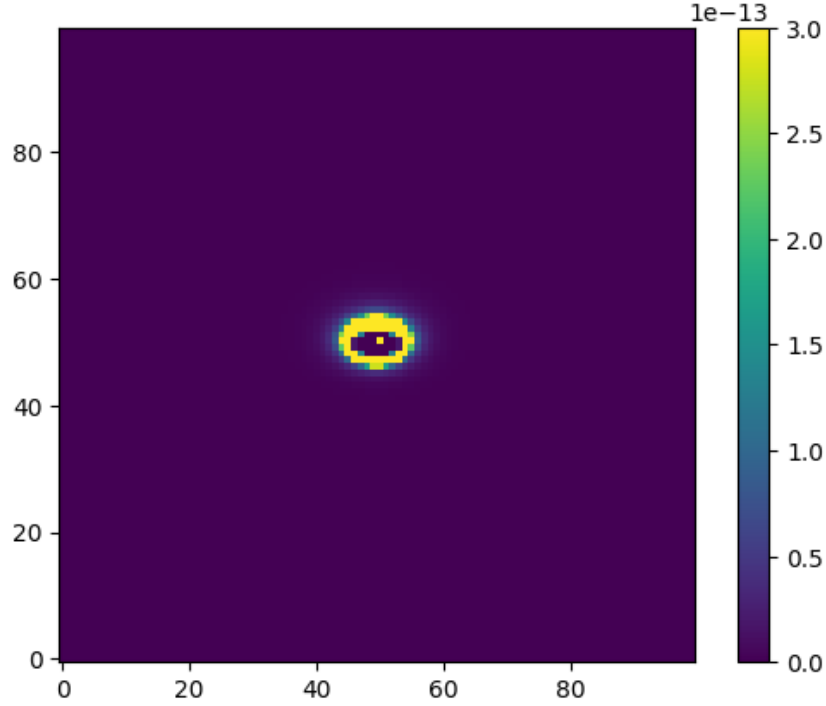


Figure 3: Synthetic Mid-Infrared Image of a Protoplanetary Disk

Modifying the coordinate axis and transposing the image matrix for better visual , our disk looks like this.

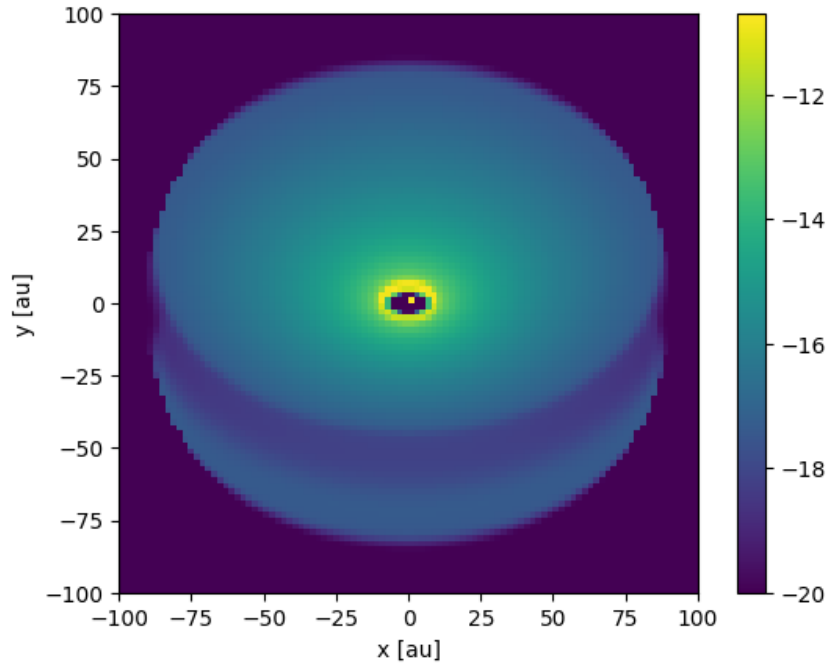


Figure 4: A re-oriented view of the disk model after transposing the image matrix and adjusting coordinate axes for improved structural clarity

Further modification with the colourbar to get a better idea of the disk temperature, we get

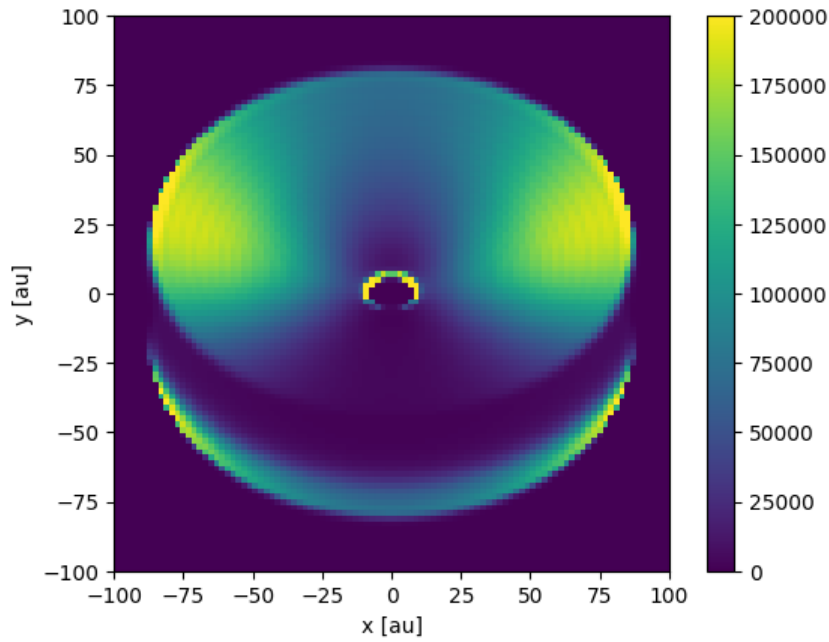
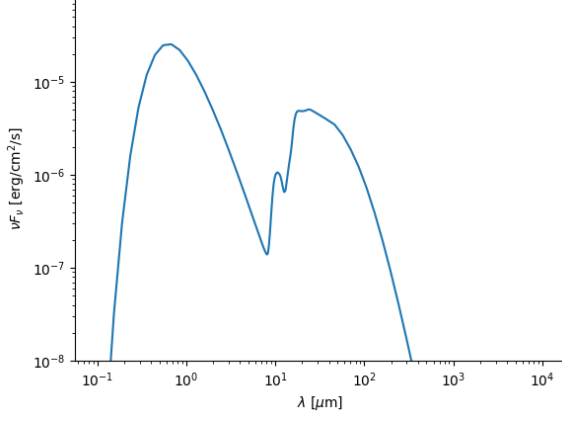


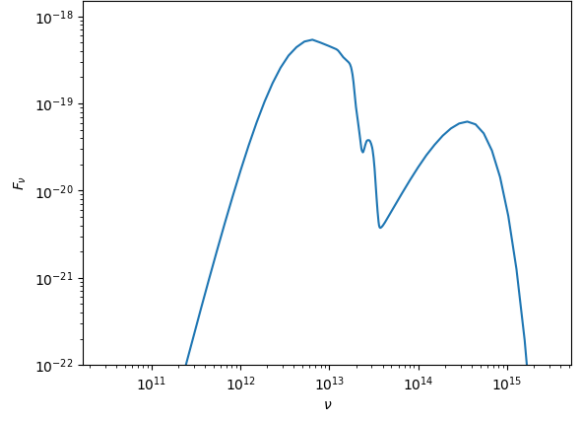
Figure 5: Detailed temperature distribution across the disk plane, with the colorbar indicating temperature in Kelvin

3.1.4 radmc3d sed incl 45 phi 30

We can also visualise the spectral energy distribution (SED) by making spectra. The output is stored in `spectrum.out`. From this same angle (inclination = 45° , $\phi = 30^\circ$), the variation of irradiance with wavelength and flux with frequency is as follows:



(a) Irradiance (νF_ν) plotted against wavelength (λ), showing the characteristic stellar peak in the optical/near-IR and the dust emission in the mid-to-far-IR



(b) Flux (F_ν) versus frequency (ν), providing an alternative representation of the system's radiative output across the electromagnetic spectrum

Figure 6: Spectral Energy Distribution (SED) of the Star-Disk System

3.2 Comparative Analysis of Geometric and Stellar Parameters

The default mass of the star in this model is $1M_{\odot}$. The default inner and outer radii of disk is $10 A.U.$ and $100 A.U.$. We vary these parameters and see how the protoplanetary disk changes. The changes are made in the `problem_setup.py` file and the dust temperature profile, image and spectra are generated.

3.2.1 Changing disk radii

We take two cases: a higher disk width and a lower disk width.

Disks	Inner Radii	Outer Radii
Default	10	100
Thick	5	200
Thin	20	60

Table 1

The dust temperature profile appears as follows:

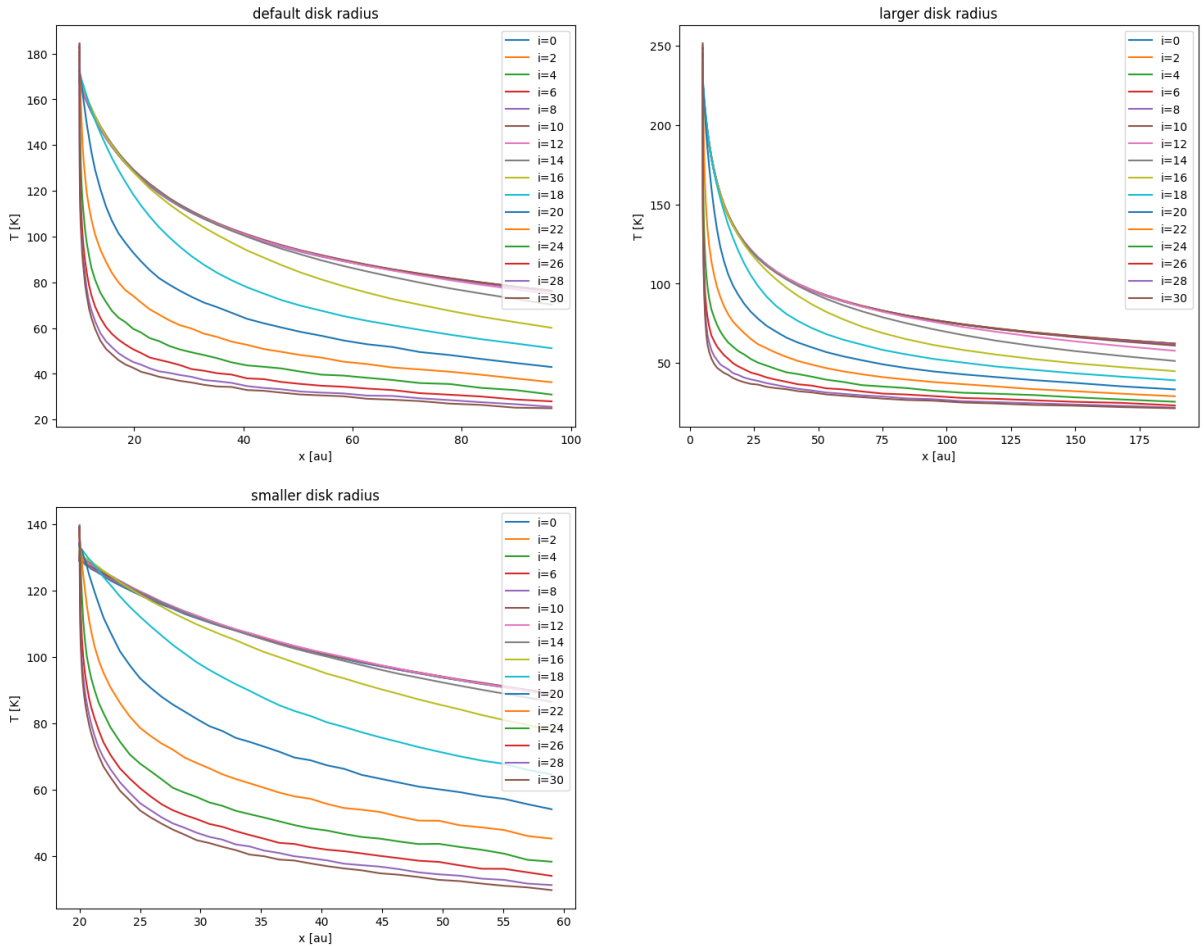


Figure 7: Comparison of radial temperature distributions for three cases: default (10–100 au), larger (5–200 au), and smaller (20–60 au) disk radii

It is clear that a thicker disc produces a steeper temperature profile (faster radial decline),

while a thinner disc results in a shallower profile (slower decline).

Visualising the star and disk system with 45° inclination and $\phi = 30^\circ$, we see:

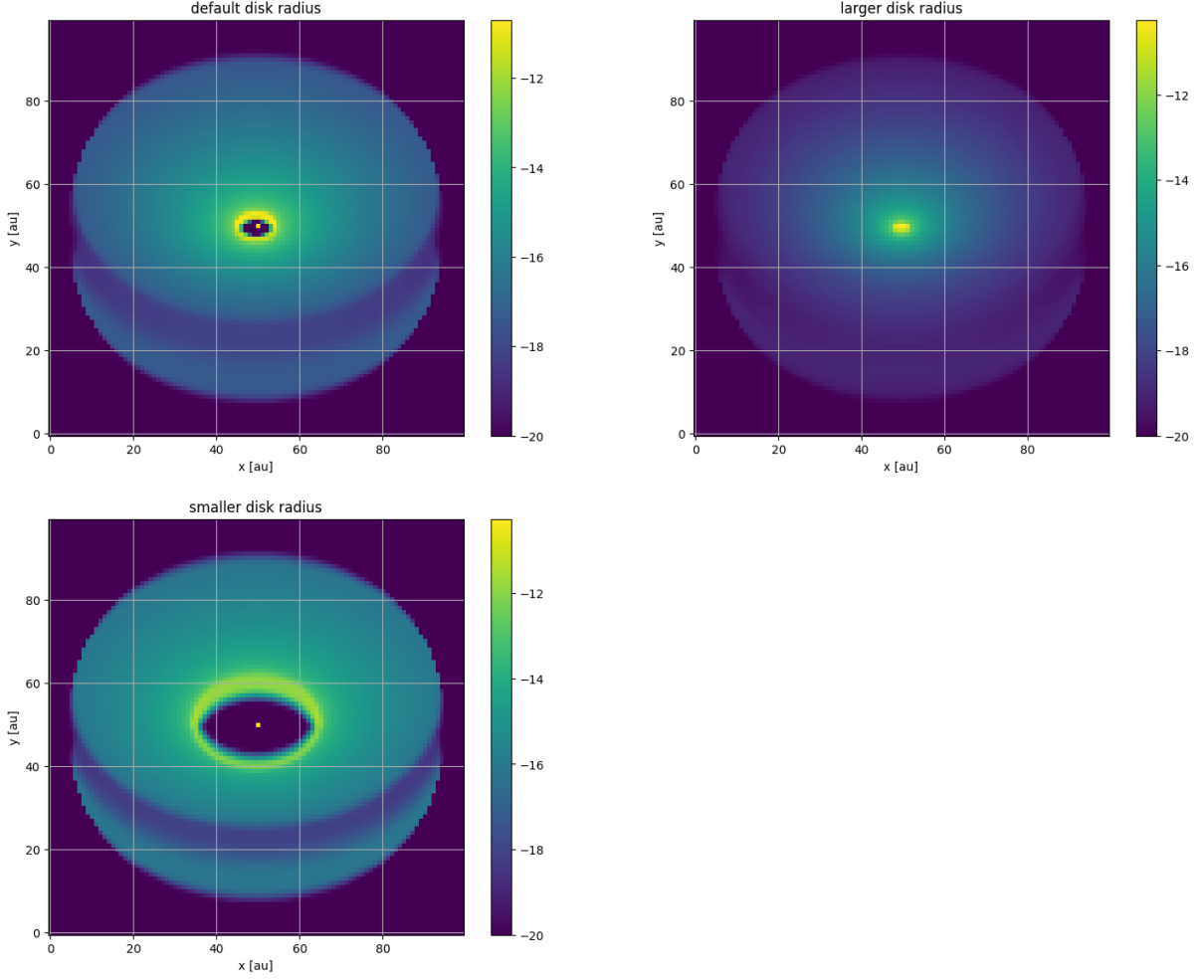


Figure 8: Simulated observations illustrating the resolvability of the central star against the disk

Intuitively enough, increasing the disk thickness makes it difficult to resolve the star separately due to higher optical depth and scattering, while a thinner disk makes a clear separation between the star and the disk.

We generate a spectra of the system as well.

The stellar irradiance and flux remain nearly constant. However, in a thicker disk, the flux spreads over a broader range of frequencies (or wavelengths), whereas a thinner disk produces a sharper, more concentrated peak.

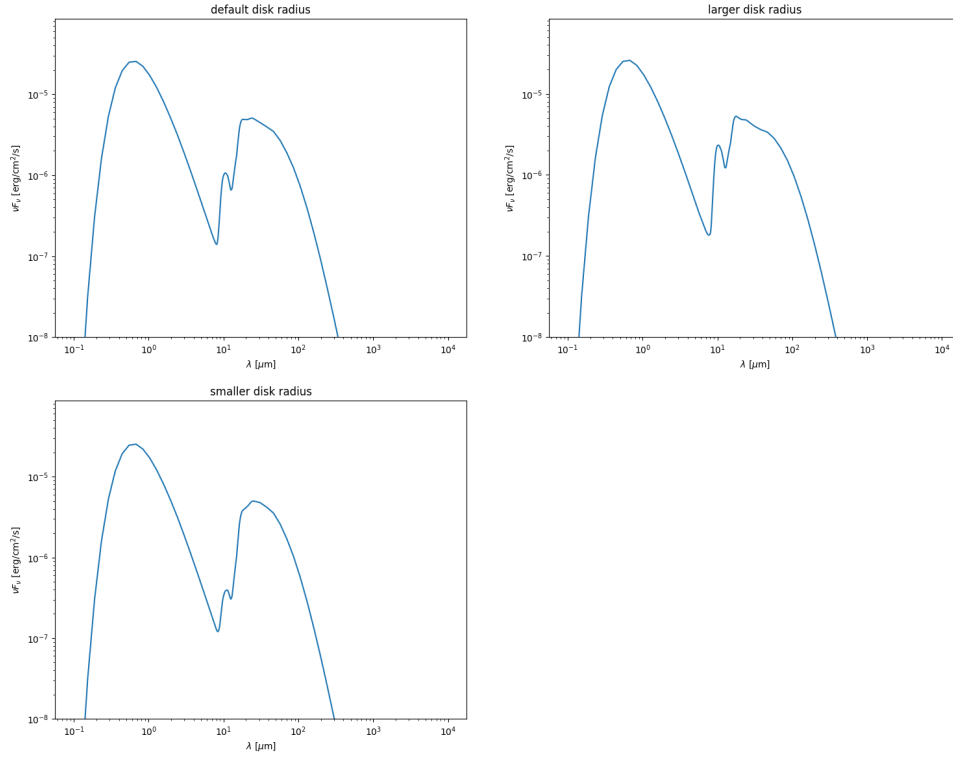


Figure 9: Comparison of integrated irradiance for different disk widths showing how it varies with the spatial extent of the dust

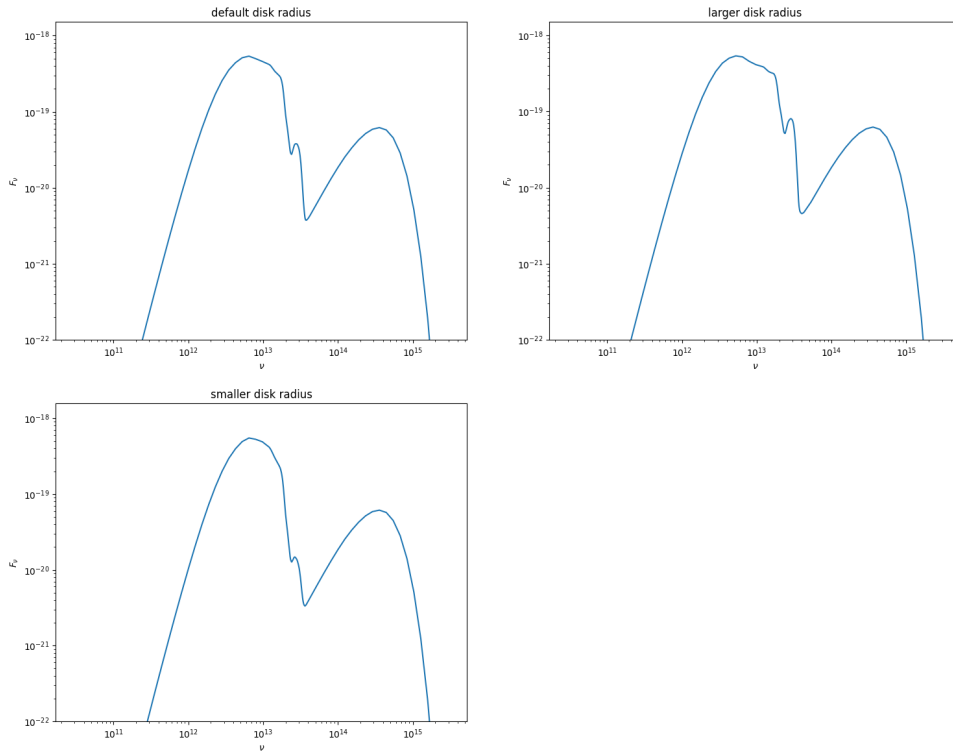


Figure 10: Comparison of spectral flux for different disk widths showing that thicker disks spread flux over a broader frequency range, while thinner disks result in more concentrated, sharper peaks

3.2.2 Changing stellar mass

To vary the star mass, again two cases are considered: a higher mass of $80 M_{\odot}$ and a lower mass of $0.1 M_{\odot}$.

The dust temperature profile is as follows:

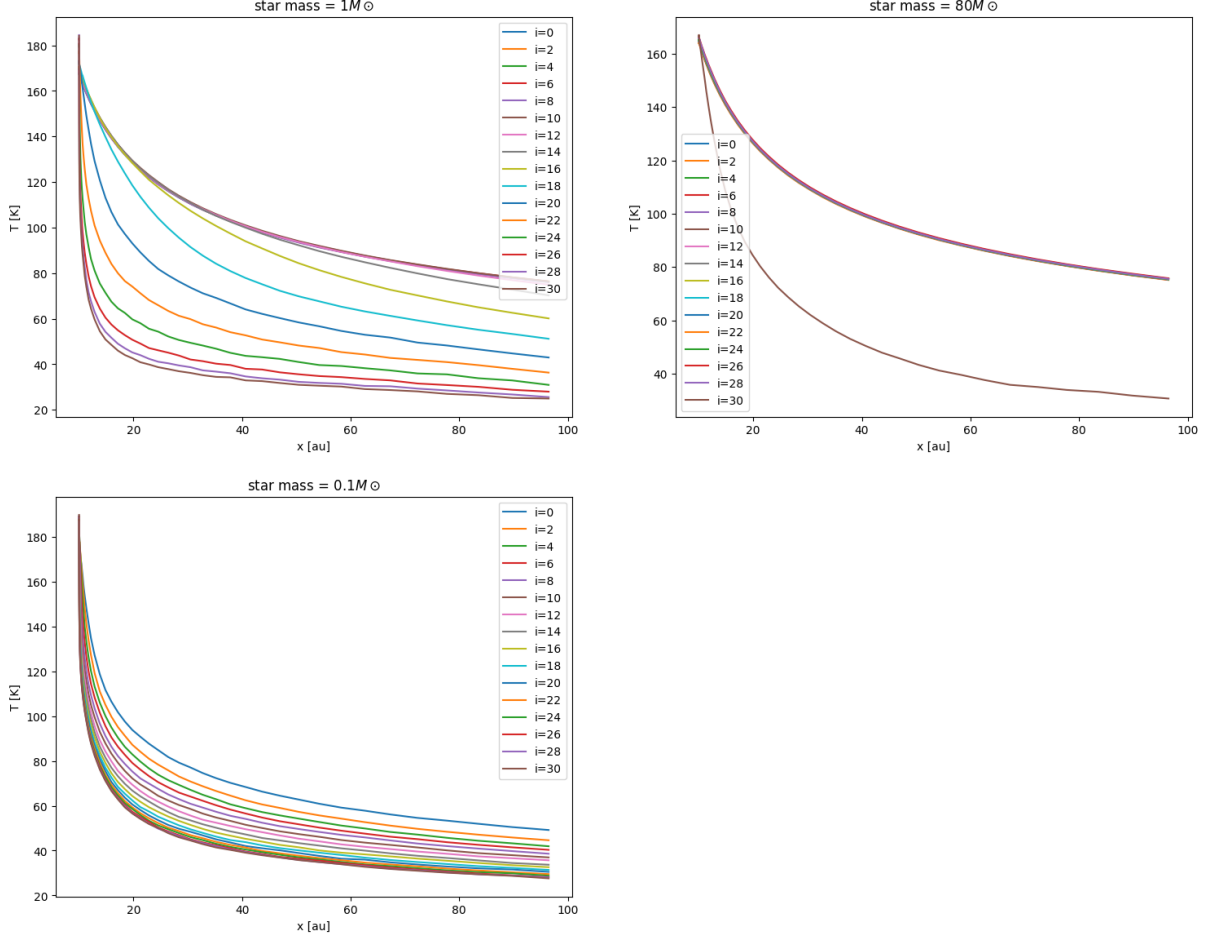


Figure 11: Temperature distributions for disks around a high-mass star ($80 M_{\odot}$) and a low-mass star ($0.1 M_{\odot}$). Higher stellar luminosity leads to a more extended heated region, whereas lower stellar mass results in a steeper temperature drop-off.

Temperature falls more steeply as the star's mass lowers. For the massive star, temperature profile overlaps for multiple layers.

Visualising the star and disk system with 45° inclination and $\phi = 30^\circ$, we see:

As the star's mass lowers, the disk becomes thicker, whereas for a high mass star, we get a very thin disk. This corresponds to the fact that a higher mass disk has more gravitational pull that can overcome the gas parcel's centrifugal barrier and make the disk more compact.

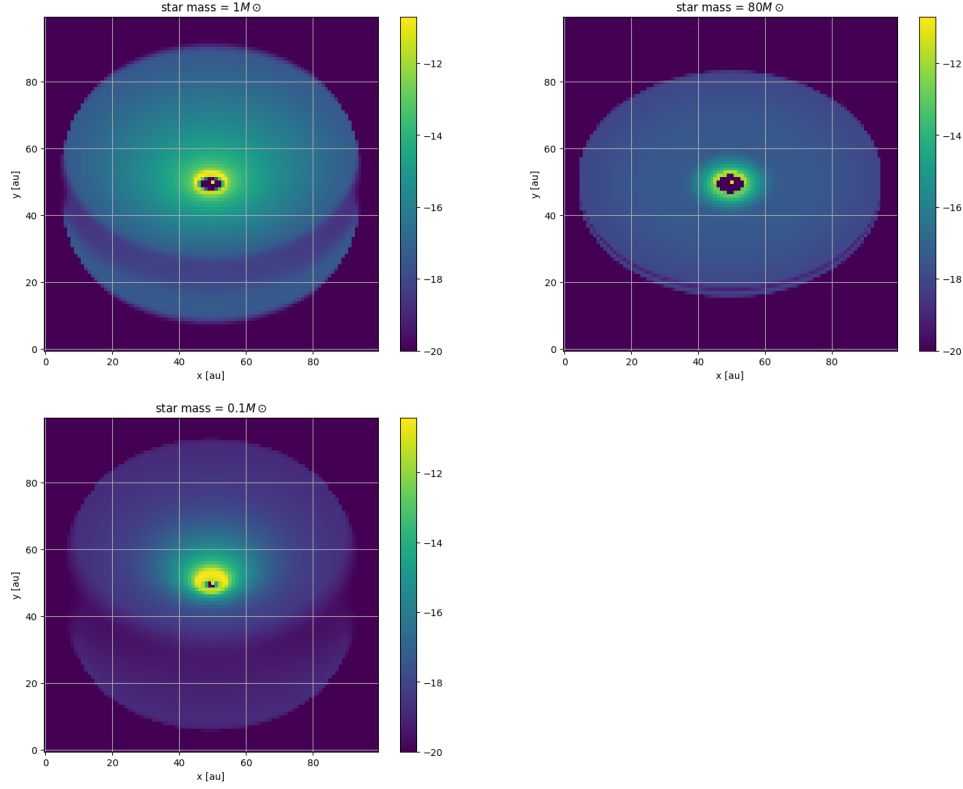


Figure 12: Disk Morphology as a function of stellar mass showing that disks around low-mass stars appear thicker, while those around high-mass stars are more compact

Again, we generate a spectra of the system.

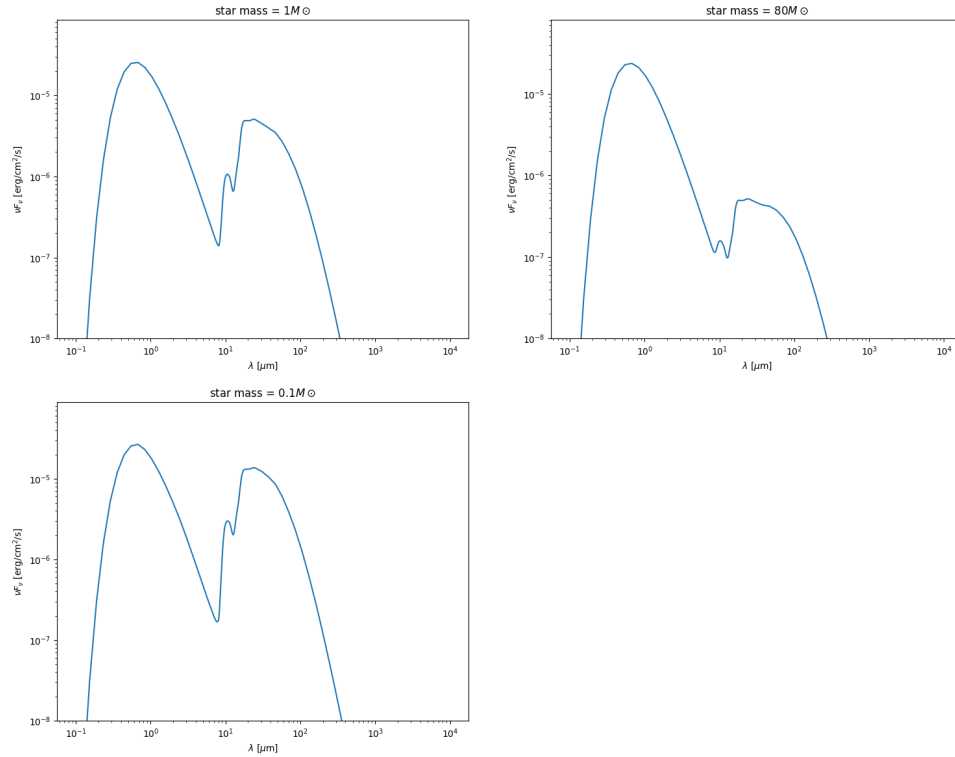


Figure 13: Shift in the peak disk irradiance as a function of stellar mass.

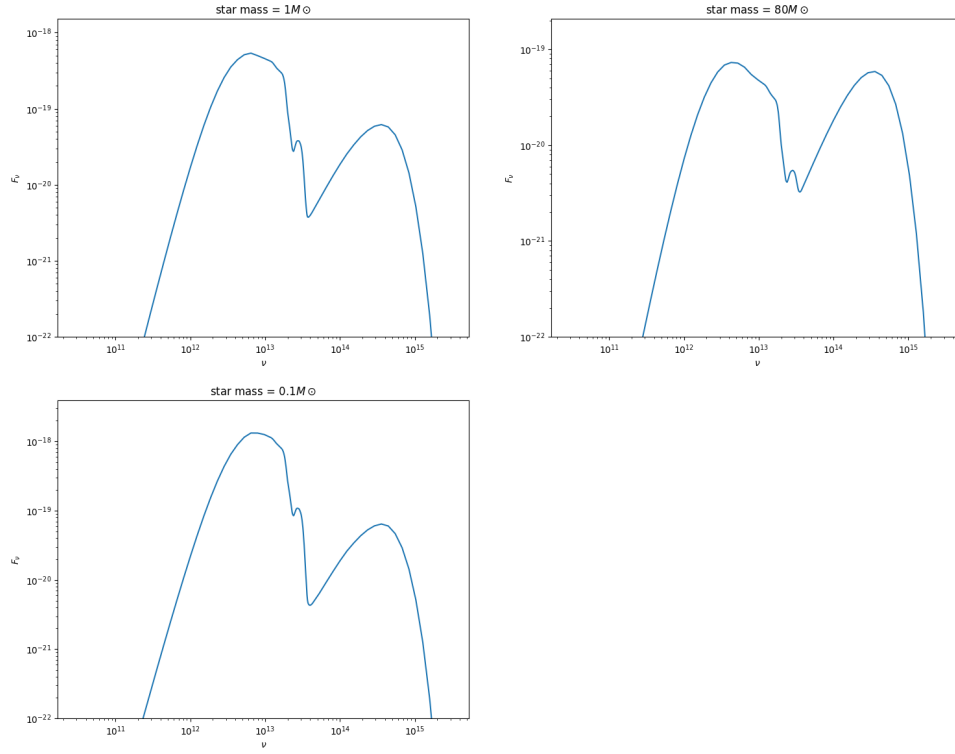


Figure 14: Frequency-dependent flux distributions illustrating that while the stellar parameters remain largely constant in the model, the disk flux rises significantly with increased stellar mass due to higher incident radiation

Peak of the disk irradiance falls with increase in stellar mass, while the flux rises. The parameters in the star however remain unaffected.

4 Modeling Warped Transition Disk Systems

We now try to compute the structure of a warped disk model. The model is written in a separate .py file and then processed using RADMC-3D. The model is built in a hierarchical way, starting from a simple 1D radial disk and extending to fully 3D, warped, and rotated configurations. It goes as follows:

1. **1D Disk(Radial structure)** - The base is a simple 1D disk model, which defines the radial grid, gas and dust surface densities, and midplane temperature. Stellar parameters such as mass, luminosity, and effective temperature are included, and the disk temperature can be computed self-consistently using a flaring angle or provided directly.
2. **2D Cylindrical Disk (Radial + Vertical)** - The 1D disk is extended into a 2D cylindrical model by introducing a vertical grid. The vertical structure is assumed to be isothermal, resulting in a Gaussian density profile. Vertical temperature variations are neglected to avoid unphysical artifacts in warped disks, and the vertical velocity structure can optionally include a Keplerian profile.
3. **3D Cylindrical Disk (Adding Azimuth)** - The cylindrical 2D model is trivially extended in the azimuthal (ϕ) direction for axisymmetric disks or can include ϕ -dependent motions such as sloshing and breathing modes. Vertical expansion due to the breathing mode can be incorporated via a specified phase-dependent factor, allowing the density and velocity structures to vary consistently in 3D.
4. **Warp Modeling** - The warp is described using vectors normal to each annulus of the disk. The warp class computes the disk geometry based on analytical models (e.g., Dullemond, Kimmig & Zanazzi 2022; Ogilvie & Latter 2013), including the local tilt, rotation, and possible precession of each disk annulus. Velocities associated with the warp—such as radial and azimuthal sloshing and vertical breathing—can be included for a physically self-consistent warped disk.
5. **Spherical 3D Disk** - The 3D cylindrical disk can be mapped onto a spherical coordinate system for radiative transfer calculations. During this conversion, each radial shell can be rotated to follow the local warp geometry. The azimuthal orientation and velocity perturbations are preserved, allowing full 3D simulations of warped, precessing, or sloshing disks suitable for synthetic observations.

We use this model for post-processing through RADMC-3D. After running the thermal Monte Carlo simulation, we generate its image from different inclinations and ϕ and also in different wavelengths. The results are as follows:

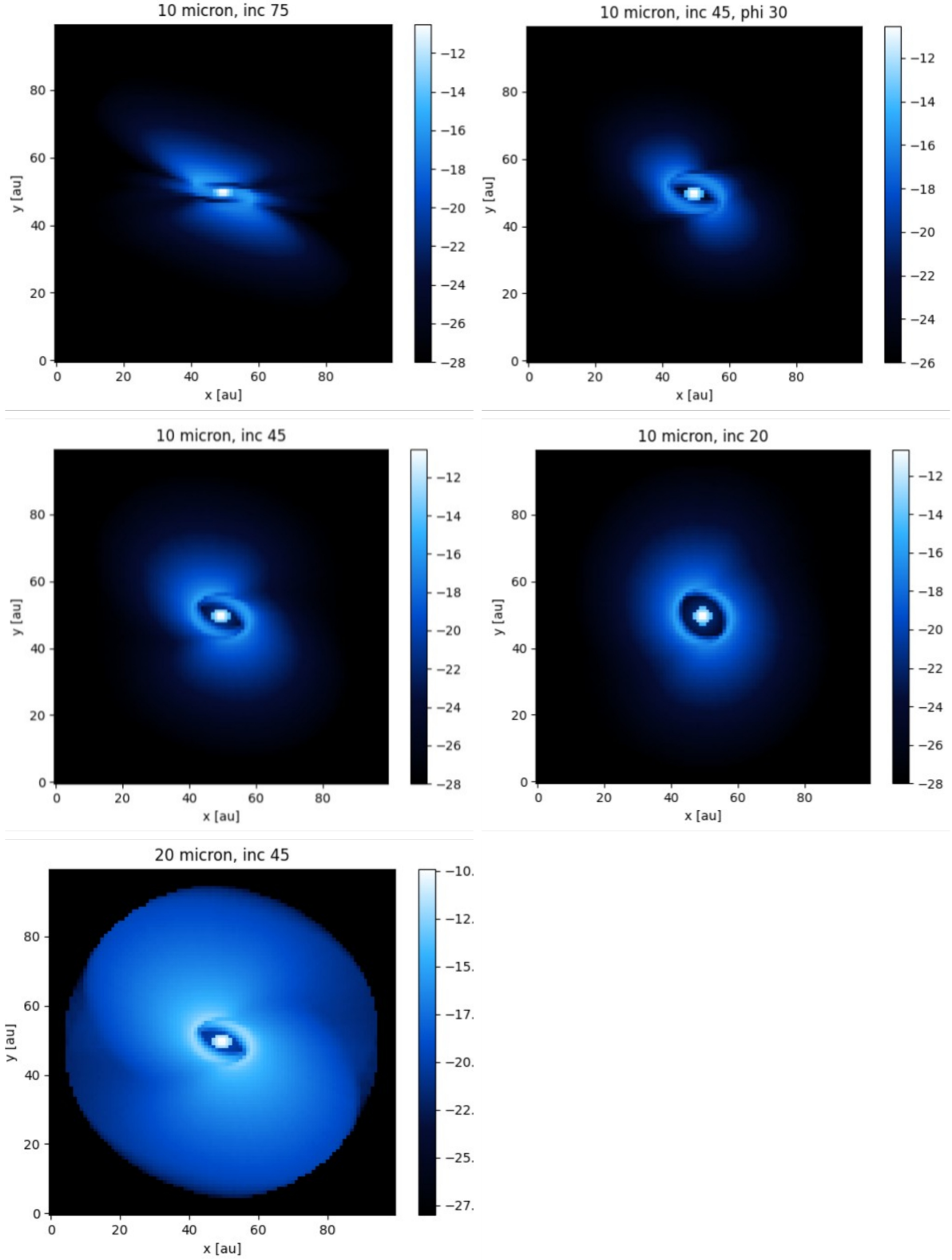


Figure 15: Synthetic 3D radiative transfer image of a warped disk model capturing the complex geometry including local tilts, rotations, and vertical perturbations, which produce characteristic asymmetries in the observed brightness distribution.

5 Conclusion

In summary, this study demonstrates that the thermal and structural evolution of protoplanetary disks is intrinsically linked to the properties of the host star and the spatial distribution of dust. By utilizing RADMC-3D to bridge the gap between theoretical models and synthetic observations, we have shown how parameters such as stellar mass and disk radii directly influence the SED and the detectability of disk features. These findings emphasize that accurately interpreting high-resolution astronomical data requires a nuanced understanding of these radiative transfer effects, as even minor variations in disk geometry can lead to significantly different observational signatures.

References

- [1] Owen, J. E. (2016). *The Origin and Evolution of Transition Discs: Successes, Problems, and Open Questions*. Publications of the Astronomical Society of Australia, 33, e005. <https://doi.org/10.1017/pasa.2016.2>
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- [3] Francis, L., van der Marel, N., Johnstone, D., Akiyama, E., Bruderer, S., Dong, R., Hashimoto, J., Liu, H. B., Muto, T., & Yang, Y. (2022). Gap opening and inner disk structure in the strongly accreting transition disk of DM Tau. *The Astronomical Journal*, 164(3), 105. <https://doi.org/10.3847/1538-3881/ac7ffb>
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